

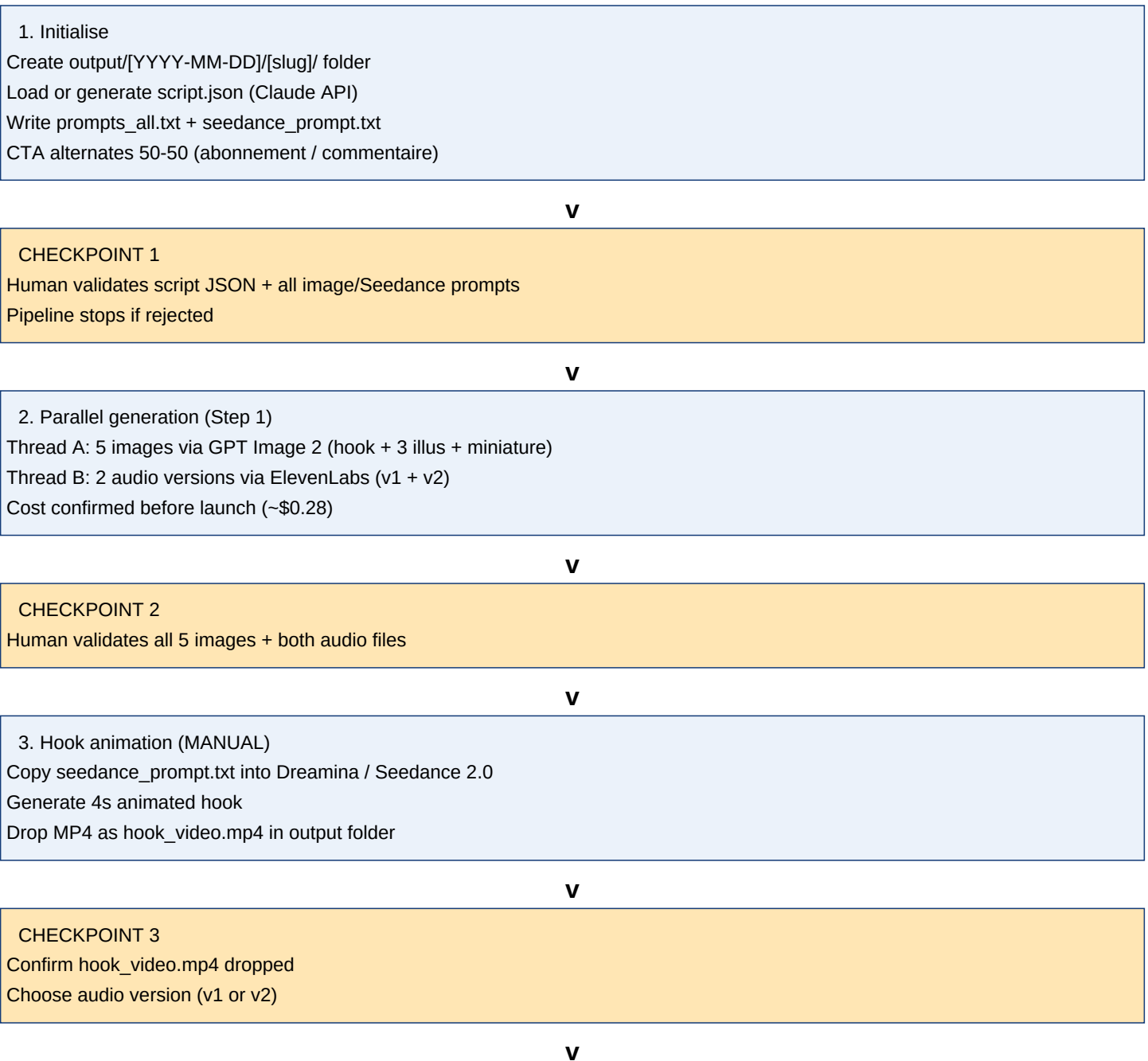
Execution Flow

curio-automation - Instagram Reels Pipeline for @curio.education

This document describes the end-to-end execution flow of the curio-automation CLI pipeline. The pipeline produces one complete Instagram Reel (28-35 seconds, 9:16) with human validation at four blocking checkpoints. Total target time: under 30 minutes per reel.

Pipeline flowchart

Entry point: python main.py with --type, --sujet (or --niveau / --matiere for competence reels).



4. Subtitles + assembly

Whisper (local) transcribes chosen audio -> subtitles.srt

FFmpeg assembles reel_final.mp4

Duration = audio length + 0.2s

One phrase per screen, ASS styled subtitles

V

CHECKPOINT 4

Human watches reel_final.mp4 and validates

V

5. Description

Claude API generates description_instagram.txt

Displayed in terminal for review

V

6. Publication (MANUAL)

Benjamin publishes manually on @curio.education

Partial rerun modes

The CLI supports relaunching individual steps on an existing output folder:

Command	Flag	Action
Regenerate images only	--only images	Requires --output-dir. Skips existing PNG files.
Regenerate audio only	--only audio	Requires --output-dir. Skips existing MP3 files.
Remount only	--assemble	Requires --output-dir. Whisper + FFmpeg. Use --audio 1 2.

Video timeline (assembly sequence)

Total reel duration = chosen audio duration + 0.2s (AUDIO_TAIL). Fixed clips total 15 seconds; the three illustrations share the remaining time in a 5:5:3 weight ratio. Pipeline errors if audio is too short (< ~16.5s).

Segment	Source file	Duration	Notes
Hook animated	hook_video.mp4	4s fixed	Dropped manually (Seedance)
Illustration 1	illus_1.png	flex (w=5)	GPT Image 2, photorealistic
Curio segment A	curio_explication.mp4	4s fixed	Reusable asset clip
Illustration 2	illus_2.png	flex (w=5)	GPT Image 2
Curio segment B	curio_explication_2.mp4	4s fixed	Reusable asset clip
Illustration 3	illus_3.png	flex (w=3)	GPT Image 2
CTA	curio_cta.mp4	3s fixed	First 2s trimmed (trim_start=2.0)

Example: audio = 24.3s -> total = 24.5s -> flex budget = 9.5s -> illus durations: 3.66s / 3.66s / 2.2s

Image reference injection

Output file	Reference inputs	Quality
hook_frame.png	curio_character_ref.png	medium (standard)
illus_1/2/3.png	style_fond_cahier + style_illustration_01/02	medium (standard)
miniature.png	miniature_exemple + illus_1.png + logo_curio.png	high

Key operational rules

- Idempotency: if an output file already exists, that step is skipped. Delete the file to regenerate.
- Visual references mandatory: generation blocked if assets/curio_reference/ is incomplete.
- API cost displayed and confirmed before every paid API call.
- Four blocking checkpoints: pipeline waits for human input (o/n) at each one.
- Whisper + FFmpeg are local and free. Hook animation (Dreamina) is manual.
- Observed API cost per reel: ~\$0.30-0.55 + Dreamina subscription (~EUR 0.42/reel amortised).